MD. Mehedi Hasan Maruf

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Career Objective

I want to teach undergraduate computer science and keep doing research in machine learning and learning analytics. My B.Sc. thesis studied which courses in a curriculum hold students back and why, so I came to teaching through the same questions a lecturer deals with every semester. I also bring several years of production software work, which I can use to make programming, database and software engineering courses closer to what students will meet in industry.

Education

Degree	Institution	Board	Result	Year
B.Sc. Engg. in CSE	Rajshahi University of Engineering & Technology (RUET)	–	CGPA 3.75 / 4.00 (Honours)	2024*
HSC (Science)	Police Lines School and College, Rangpur	Dinajpur	GPA 5.00 / 5.00	2020
SSC (Science)	Police Lines School and College, Rangpur	Dinajpur	GPA 5.00 / 5.00	2018

*Examination year 2024 (session 2020–21). Because of the COVID-19 delay, classes ran Mar 2022 – Aug 2026; the degree was awarded on 11 Aug 2026.

- Class of 178; 160 credits; BAETE-accredited program (Washington Accord). CGPA rose from 3.62 after the first year, with semester GPAs of 3.96, 3.88 and 3.84 in three consecutive semesters.
- A+ in Data Structures, Database Systems, Operating Systems, Theory of Computation, Security and Privacy (theory and lab), Applied Statistics and Queuing Theory, and the B.Sc. Thesis. A in Algorithm Analysis and Design, Artificial Intelligence (lab A+), Data Mining and Computer Networks; A− in Machine Learning.

Thesis and Research

TRACE: Curriculum Bottleneck Detection and Causal Validation *Jul 2025 – Aug 2026*

B.Sc. Thesis, Dept. of CSE, RUET

- Started from my own class, where about a third failed one first-year mathematics course (about 4% fail a typical course). Does such a course hold students back, or only show who was already behind?
- Built TRACE, which ranks courses by prerequisite-graph position (NetworkX, GAT), a Bayesian fail-rate model, and student-feedback sentiment (DeBERTa, VADER).
- Reached F-measure 0.75 and NDCG@10 0.81 on a synthetic curriculum with planted bottlenecks; on a local university's curriculum, a domain expert agreed with 8 of its top 10 courses.
- Used causal inference to test what drives failure in a flagged module: on 23,369 OULAD students, DoWhy estimated that high assignment engagement raised held-out pass rates by 19.9 points ($p < 0.01$).

Publications

- *When Agreement Misleads: Auditing Course Shortlists in Curriculum Analytics.* Submitted to the International Conference on Learning Analytics & Knowledge (LAK 2027). Under review.

Research interests: machine learning and NLP, causal inference, graph learning, learning analytics, evaluation of LLM and retrieval-augmented systems.

Teaching and Mentoring

- **Competitive Programming Mentor**, RUET Advanced Programming Lab (RAPL), Aug – Nov 2023. Taught number theory and graph traversal to junior students, prepared the problem sets and ran biweekly contests.
- **Class Representative**, 4th year odd semester, Dept. of CSE, RUET. Handled schedules, exams and students' problems between my batch and the course adviser.
- **Team Representative**, Brain Station 23 industrial training, Mar 2025. Represented a 15-person trainee cohort.

Courses I can teach

Programming and core CS: Structured Programming (C), Object-Oriented Programming, Data Structures, Algorithms, Discrete Mathematics, Theory of Computation, Operating Systems
Data and AI: Database Systems, Artificial Intelligence, Machine Learning, Data Mining, Applied Statistics
Systems and software: Software Engineering, Web Engineering, Computer Networks, Computer Security
Labs: all of the above, plus project and capstone supervision on web, backend and AI applications

Professional Experience

Software Engineer II
Jan 2025 – Present

Aamira Software Solutions, Kokomo, Indiana, USA (remote)

- Raised client response rates by 30% by leading development of a customer relationship management (CRM) system, and made the application 60% faster by optimizing its Supabase queries.
- Cut API response time by 40% by designing a TypeScript microservice backend.

Junior Software Engineer
Nov 2022 – Apr 2024

Code Studio, Rajshahi, Bangladesh

- Kept 12 Linux servers at 99.8% uptime, cut deployment time by 40% by reworking the CI/CD pipelines, and moved 5 client environments onto Docker.
- Wrote JavaScript/TypeScript and Python (Django) backends used by thousands of people a day.

AI Code Reviewer / RLHF Evaluator
Oct 2024 – Jan 2026

Outlier (remote)

- Reviewed code for AI systems used by 50,000+ people a day; evaluated and annotated large language model responses for correctness, consistency and safety.

Industrial Trainee
Mar 2025

Brain Station 23, Dhaka, Bangladesh. Built StudentAI (see Projects).

Selected Projects

- **Omnious** (2025 – present): a visual debugging tool that draws a program's code graph, animates execution traces and lets you ask an AI assistant about errors. Fastify, tRPC, Next.js, PostgreSQL with pgvector.
- **StudentAI** (2025): document Q&A, YouTube lecture analysis and quiz generation for students, with semantic retrieval and conversation memory. Takes PDFs, images and video.

Skills

AI/ML: PyTorch, Hugging Face Transformers, graph neural networks, LangChain, RAG, DoWhy (causal inference), Bayesian modeling
Languages: C, C++, Python, TypeScript, JavaScript, SQL
Web and systems: NestJS, Next.js (React), Django, FastAPI, PostgreSQL, MySQL, MongoDB, Docker, GitHub Actions, Linux server administration

Awards and Certifications

- 4th place, RUET CodeSmash team programming contest (Mar 2023)
- 4th place, RUET CodeSpark individual programming contest (Oct 2022)
- Digital Accessibility Training, a2i, Government of Bangladesh (Oct 2025)
- PHP (Laravel), EDGE Project, ICT Division (Feb 2025)

Personal Information

Date of birth:	03 September 2002	Nationality:	Bangladeshi
Father's name:	MD ABDUL KHALEQUE MONDOL	Mother's name:	MOST MASUMA BEGUM
Languages:	Bangla, English	Permanent address:	Gaibandha

References

Prof. Dr. Md. Shahid Uz Zaman
Professor and Head, Dept. of CSE, RUET
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Taught Data Structures; thesis examiner

Mohiuddin Ahmed
Assistant Professor (on study leave), Dept. of CSE, RUET
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Course adviser for four years

Dr. Md. Nour Hossain
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Research advisor; current employer